

2021	X	05	1030	RM-460	(E)
CROP SCIENCE PAPER - I (B-4)					
Time : 2 Hours		8 Pages		Max. Marks : 40	

General Instructions to Moderators and Examiners :

- (1) They should adhere to the model answers and scheme of marking.
- (2) Importance should be given to the value points rather than mere description.
- (3) Equally, correct points, which are not mentioned in the model answers, should be given credit.

1. (A) Fill in the blanks :

- | | |
|---------------------------|---|
| (a) – Maize | 1 |
| (b) – Soil - Fertility | 1 |
| (c) – 20 to 37°C | 1 |
| (d) – Helianthus amunsus | 1 |
| (e) – Tiliaceae | 1 |
| (f) – CK-60 A and IS – 84 | 1 |

(B) Define : (Any Two)

- | | |
|--|---|
| (a) Seed : Any material used for planting and propagation, it is in the form of grains, seedlings, tubers, bulbs, rhizomes, cuttings and any other vegetatively propagated material is called as seed. | 1 |
| (b) Monoculture : | 1 |
| The practise of taking one crop on same piece of land year after year. | |
| (c) Tillage : | 1 |
| Tillage of soil consists of breaking the hard and compact surface of soil to a certain depth and other operation which are followed for bringing the soil in an ideal condition for proper growth of the plants. | |

2. Short notes on : (Any Four)

(a) **Retting and extraction of Fibres in Sunnhemp :** 2

- (i) The bundles are retted in standing water to 60 to 90cm. depth.
- (ii) Retting is done for 2 to 3 weeks.
- (iii) Fibres are then removed by hand.
- (iv) Extracted fibres are washed in clean water.
- (v) Dried for 2 to 3 days in hanging position.

(b) **Ergot disease of Bajra :** 2

- (i) It is a serious disease of bajra.
- (ii) A pinkish sugary substance oozes out from the flower heads and drops on the lower leaves.
- (iii) Affected ovaries are transformed into dark brown elongated sclerotia.
- (iv) Seeds should be dipped in 20% common salt solution for its control.

(c) **Tur Pod Borer :** 2

- (i) It is a most sever pest of tur crop.
- (ii) The larva bores into the pod and feeds on developing grains.

C.M. :

- (i) Dusting of 1.3% Lindane powder at the rate of 15 kg./ha.
- (ii) Spraying of 0.06% Endosul fan at 20 days interval.

(d) **Characteristics of Good Cropping Scheme :** 2

- (i) It should include legume crops which help in crop rotation and maintaining soil fertility.
- (ii) It should engaged labours throughout the year.
- (iii) It should provide food to family and fodder to cattle.
- (iv) It should give cash to farmers.
- (v) It should reduced cost of cultivation.

(e) **Seed Treatment in Jowar :** 2

- (i) Treatment of 300 mesh fine sulphur powder.
- (ii) Treatment of carbofuran.
- (iii) Treatment of 30% brine solution.

Above treatments, their objects and proportion of chemicals to be used.

(f) **Economic uses of Castor :**

2

- (i) Castor oil is non edible.
- (ii) It has non drying and nonfreezing quality at low temperature.
- (iii) It is used in the manufacture of high grade lubricating oil.
- (iv) It is a laxacative and is used as medicine.
- (v) Oil cake has vermicidal properties.
- (vi) The stalks are used for making paper pulp.
- (vii) Oil contains is 46 to 47%.

3. Attempt **any four** of the following :

8

- (a) (i) Can be grown on all kinds of soils and important constituent of crop rotation.

2

- (ii) Crop like groundnut not only maintains soil fertility but also reduce cost of land preparation of next crop.

- (iii) Valuable cash crop and brings ready cash to farmer.

- (iv) Source of foreign exchange.

- (v) Provide raw material for many industries.

- (vi) Edible oil cakes are used as cattle feed.

- (vii) Crops like castor, sesamum can be grown on bunds and give additional income to the farmer.

(b) **Aspects of Farm Management :**

2

- (i) Choice of an enterprises.

- (ii) Selection of a farm/

- (iii) Preparing a suitable layout.

- (iv) Formulation of cropping scheme and calender of operations.

- (v) Equipping the farm with necessary inputs.

- (vi) Management of farm labour.

- (vii) Farm Records

(c) **Harvesting of Paddy :**

2

- (i) Paddy crop matures within 120, 130 and 150 days i.e. early, medium and late varieties respectively.
- (ii) Harvesting is done when earheads are partially ripened and tillers are yellowish in colour i.e. semimatured.
- (iii) Delay in harvesting may caused shedding of grains partially in case of early varieties.
- (iv) Harvesting at proper stage is very important in case of paddy.
- (v) Plants are cut close to ground level.
- (vi) They are allowed to dry a about 3 to 4 days in the same field.
- (vii) Dried plants are taken to threshing yard.
- (viii) Threshing is done by beating the material with sticks.
- (ix) Grains are separated by winnowing.
- (x) Now-a-days threshing is done with the help of power thresher.

Yield :

- (i) Rainfed local var - 10 to 15 qt/ha.
- (ii) Irrigated local var - 20 to 25qt/ha.
- (iii) Rainfed local high yielding var - 25 to 30 qt/ha.
- (iv) Irrigated high yielding var - 40 to 50 qt.ha.

(d) **Improvement of Labour efficiency :**

2

- (i) Use of efficient implement and machineries.
- (ii) Provide drinking water and food at working place.
- (iii) Have a large sized field.
- (iv) Make a regular payment of wages.
- (v) Select suitable man as Mokadam for supervision.

(e) **Harvesting and Yield of Niger :**

2

- (i) Niger crop matures in 4 to 5 months.
- (ii) It is harvested when the leaves are dry and heads turn blackish.
- (iii) The crop is cut close to the ground and tied into bundles.
- (iv) They are threshed by beating with sticks.
- (v) Pure crop yield is 3 to 4 quintal/ha.
- (vi) Mixed crop yield is 1 quintal/ha.

(f) **Cultivation Practices of Paragrass :**

2

- (i) Land is ploughed and harrowed.
- (ii) 25 to 30 tones F.Y.M./ha. is mixed into the soil during the preparation of land.
- (iii) Rooted sets are used for planting at 20 × 20 cm. distance.
- (iv) First cutting is para grass is available in 2.5 to 3 months after planting.
- (v) Other cuttings are available at six weeks interval.
- (vi) In a year 8 to 10 cuttings are obtained.
- (vii) Yield of green para gras fodder in a year is a bout 100 to 125 tons/ha.

4. (A) **Maize : (Any Two)**

4

(a) **Preparation of Land :**

2

- (i) The land is ploughed and brought to fine tilth by giving two to three harrowing.
- (ii) The seed bed should be free from weeds and stubbles.
- (iii) FYM is added and thoroughly mixed with soil by final harrowing.
- (iv) Planking is done to make compact seed bed.

(b) **Manuring and Fertilizers :**

2

Manures : 12 – 15 tons of FYM/ha. is added at the time of land preparation. <https://www.maharashtrastudy.com>

Fertilizers : dose :

For local varieties - 30 kg/ha.

For hybrid varieties - 75 to 100 kg N;

50 to 75 kg/ P_2O_5 , 50 to 75 kg. K_2O /ha.

(c) **Economic Uses :**

2

- (i) Maize is an important cereal as well as fodder crop.
- (ii) Maize is used as a staple food in North India.
- (iii) Some varieties are consumed as popgrain.
- (iv) In dough stage, they are baked on fibre and the grain is eaten.
- (v) Provides sweet succulent fodder.

(B) **Groundnut : (Any Two)**

4

(a) **Ecology :**

2

- (i) It is tropical and subtropical crop.
- (ii) Frost and drought condition are harmful.
- (iii) Annual rainfall 675 to 1000 mm.
- (iv) It can be grown on sandy loam, loam and well drained soil.

(b) **Seedrate and Spacing :**

2

(i) Seedrate - 70 to 80 kg/ha.

(ii) Spacing - 25 to 30 cm.

(c) **Varieties :**

2

(i) JL - 24

(ii) AK - 12 - 24

(iii) Ak - 10

(iv) Kopargaon No-3

(v) Faizpur 1-5

(vi) Karad 4-11

(vii) TG - 17

OR [For 4(A) and (B)]

4. Give tabular information (Each correct answer carries $\frac{1}{2}$ mark)

8

Name of Crop	Botanical Name	Seedrate kg/ha.	Variety (Any one)	Yield kg/ha.	
Kidney Bean	<i>Phaseolus oconitifolius</i>	18 to 20	Local Improved	400 to 450	2
Sesamum	<i>Sesamum indicum</i>	12.5 (Broad) 2.5 to 7.5D	No : 85, 8, 128 Phule-1, JLT - 7 TMV-1	250 to 375 (R) 250 to 550 (K) 700 to 800	2
Deccan Hemp	<i>Hibiscus species</i>	30 - 35	HC-582, HC-583, HC-867, HC-4288	Fibre 100 to 200 Seed-600 to 800	2
Green gram	<i>Phaseolus aureus</i>	18 to 20	Kopargaon, T-1, Pusa-Baisakhi, J-781	500 to 550 550 to 600	2

5. **Irrigated Cotton : (Any Four)**

8

(a) **Ecology :**

2

- (i) Cotton is subtropical crop.
- (ii) Temperature rang between 21 to 27°C.
- (iii) Rainfal 625 to 1.250 mm.
- (iv) Sandy soils of Rajasthan to ferfile black soils of Maharashtra.

(b) **Sowing Time and Sowing Method :**

2

- (i) Decan region - March end to early Apirl
- (ii) Marathwada - 1st week of May.
- (iii) Vidarbha - 15 to 30th May

Sowing Method : Sowing methods followed are –

Drilling, Dibbling and Transplanting

(c) **Pests and Disease (Names only)**

Pests :

Before flowering : Aphids, Jassides, Thrips, White Files

After flowering : Spotted bollworm, American bollworm, Pink bollworm

Disease :

2

- (i) Cotton wilt
- (ii) Dahiya
- (iii) Anthracnose
- (iv) Black arm

(d) **Varities : (Any Four)**

2

- (i) Deshi : G-6, G-12, G-22, Eknath, Rohini
- (ii) Amerian : Buri-147, Buri-1007, Buri-394, MCU-5
- (iii) Hybrid : H4, Varlaxmi, NHH 44, DCH-32, Godavari
- (iv) Egyptian : Sujata

(e) Picking and Yield :

2

- (i) Harvesting is done in 3 to 4 picking at 15 to 20 days interval.
- (ii) Middle picking are heavier and best quality.
- (iii) Last picking is of poor quality and should be kept separately.
- (iv) Period remains for 8 to 10 weeks.
- (v) Yield :
 - (1) Deshi - 4 to 6 qt./ha.
 - (2) American - 10 to 12 qt./ha.
 - (3) Hybrid (Rainfed) - 12 to 20 qt./ha.
 - (4) Hybrid (Irrigated) - 20 to 30 qt./ha.

(f) Quality Aspects :

2

- (i) Staple length
 - (ii) Fineness
 - (iii) Maturity
 - (iv) Strength
 - (v) Ginning percentage
-

<https://www.maharashtrastudy.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

RM-460/8

<https://www.maharashtrastudy.com>